Final Exam

Graded

Student

Tianqi Yin

Total Points

44 / 45 pts

Question 1

Overall Grades **44** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 1 pt** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Tianqi Yin	tq144	Tianqi Yin

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Tianqi Yin

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	4	5	5	44

Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$R_3 \leftrightarrow R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right]$$

$R_3 \leftarrow R_3 - 2R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$R_2 \leftarrow (-\frac{1}{2})R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$R_3 \leftarrow R_3 + 3R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & -\frac{5}{2} \end{array}\right]$$

$R_3 \leftarrow 2R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$R_2 \leftarrow R_2 - \frac{3}{2}R_3$

$R_1 \leftarrow R_1 - 2R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

Hence, $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1)

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$R_1 \leftarrow \frac{1}{2} R_1$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$R_2 \leftarrow R_2 + 5R_1$

$R_4 \leftarrow R_4 - R_1$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$R_2 \leftarrow \frac{1}{10} R_2$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$R_3 \leftarrow R_3 - R_2$

$R_4 \leftarrow R_4 - R_2$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_1 \leftarrow R_1 - 2R_2$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, rank $(A) = 2$

Since pivot entry lies in first two Columns.

thus, the basis of col (A) is $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) by (1)

$$A \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and since we did not switch rows when operating on A and since the pivot entry lies in first two row.

thus, the basis of row (A) is $\left\{ [2\ 4\ 2], [-5\ 0\ 5] \right\}$

(3) by (1) $A \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Hence, the solution to $A\vec{x} = \vec{0}$ is $\vec{x} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Hence, the basis of null (A) is $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = A\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$, $T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = A\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix}$, $T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = A\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$

by column method of matrix multiplication

$$A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right]$$

as calculated, $\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$

$R_1 \leftrightarrow R_2$

$$\sim \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\-1&-1&-1&0&0&1\end{array}\right]$$

Hence, $A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}^{-1}$

$R_3 \leftarrow R_3 + R_1$

$$\sim \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&-1&-1&0&1&1\end{array}\right]$$

$$= \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

$R_3 \leftarrow R_3 + R_2$

$$\sim \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&-1&1&1&1\end{array}\right]$$

$$= \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$$

$R_3 \leftarrow (-1)R_3$

$$\sim \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right]$$

Hence, $\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$ is invertible, and

$$\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

Problem 4. [5?] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ $\quad \det(\lambda I - A) = \begin{vmatrix} \lambda-1 & -2 & 0 \\ 0 & \lambda-2 & 0 \\ 2 & 2 & \lambda-1 \end{vmatrix} = (\lambda-1)^2(\lambda-2)$

Let $\det(\lambda I - A) = 0$, $\quad$ algebraic multiplicity of $\lambda=1$ is 2

$\lambda = 1$ or 2

When $\lambda = 1$, $\lambda I - A = \begin{bmatrix} 0 & -2 & 0 \\ 0 & -1 & 0 \\ 2 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad (\lambda I - A)\vec{v} = \vec{0}$

$\vec{v} = t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

When $\lambda = 2$, $\lambda I - A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 2 & 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{1}{6} \\ 0 & 0 & 0 \end{bmatrix}$ $\quad (\lambda I - A)\vec{v} = \vec{0}$

$\vec{v} = t\begin{bmatrix} -\frac{1}{3} \\ \frac{1}{6} \\ 1 \end{bmatrix}$

Hence, when $\lambda = 1$, $\dim\left(\text{Null}(\lambda I - A)\right) = 1$.

Hence, geometric multiplicity of $\lambda = 1$ is 1

Therefore, Jordan canonical form of A is $J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$P = \begin{bmatrix} 0 & & -\frac{1}{3} \\ 0 & \vec{v}_2 & \frac{1}{6} \\ 1 & & 1 \end{bmatrix}$ $\quad AJ = PJ \Rightarrow (A-2I)\vec{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\Rightarrow \vec{v}_2 = t\begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$

Hence, $P = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{6} \\ 1 & 1 & 1 \end{bmatrix}$, by calculation $P^{-1} = \begin{bmatrix} 2 & 2 & 1 \\ -2 & 4 & 0 \\ 0 & -6 & 0 \end{bmatrix}$

$A = PJP^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{6} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 4 & 0 \\ 0 & -6 & 0 \end{bmatrix}$

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$f(x,y,z) = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Let $A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$

$\det(\lambda I - A) = \begin{vmatrix} \lambda-3 & 0 & -1 \\ 0 & \lambda-3 & 1 \\ -1 & 1 & \lambda-4 \end{vmatrix} = (\lambda-3)(\lambda-2)(\lambda-5)$ $\quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Let $\det(\lambda I - A) = 0$, $\lambda = 3$ or 2 or 5

when $\lambda = 3$ $\quad \lambda I - A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad (\lambda I - A)\vec{v} = \vec{0}$

$\vec{v} = t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

normalize, $\vec{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$

when $\lambda = 2$ $\quad \lambda I - A = \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad (\lambda I - A)\vec{v} = \vec{0}$

$\vec{v} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

normalize $\vec{v}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$

When $\lambda = 5$ $\quad \lambda I - A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$ $\quad (\lambda I - A)\vec{v} = \vec{0}$

$\vec{v} = t \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$

normalize $\vec{v}_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$

Hence, $P = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{bmatrix}$ $\quad P^{-1} = P^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}$

Hence, $A = PDP^T$ $\quad f(x,y,z) = \begin{bmatrix} x & y & z \end{bmatrix} P D P^T \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y \\ -\frac{1}{\sqrt{3}}x + \frac{1}{\sqrt{3}}y + \frac{1}{\sqrt{3}}z \\ \frac{1}{\sqrt{6}}x - \frac{1}{\sqrt{6}}y + \frac{2}{\sqrt{6}}z \end{bmatrix}$ $\quad$ then $f(x,y,z) = \vec{u}^T D \vec{u} = 3u_1^2 + 2u_2^2 + 5u_3^2$

Let $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = P^T \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\left[\begin{array}{ccc|c} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{w} \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

Hence $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

$\vec{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$.

$$\left[\begin{array}{ccc|c} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{w} \end{array}\right] = \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\sim \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$R_3 \leftarrow R_3 + R_1$

$$\sim \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$R_3 \leftarrow R_3 - \frac{1}{2}R_2$

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & \frac{3}{2} & \frac{3}{2} \end{array}\right]$$

$R_1 \leftarrow (\frac{1}{2})R_1$

$R_2 \leftarrow (\frac{1}{2})R_2$

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & 4 \\ 0 & 1 & -\frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$R_3 \leftarrow \frac{2}{3}R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & 4 \\ 0 & 1 & -\frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1 - \frac{1}{2}R_3$

$R_2 \leftarrow R_2 + \frac{1}{2}R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

4/5

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Let $A = \begin{bmatrix}0 & 3\\ 1 & 0\\ 1 & 2\\ 0 & -1\end{bmatrix}$

$R_1 \leftrightarrow R_2$ $\sim \begin{bmatrix}1 & 0\\ 0 & 3\\ 1 & 2\\ 0 & -1\end{bmatrix}$

$R_3 \leftarrow R_3 - R_1$ $\sim \begin{bmatrix}1 & 0\\ 0 & 3\\ 0 & 2\\ 0 & -1\end{bmatrix}$

$R_4 \leftarrow (-1)R_4$ $\sim \begin{bmatrix}1 & 0\\ 0 & 3\\ 0 & 2\\ 0 & 1\end{bmatrix}$

Hence, pivot entry in 2nd col

$R_4 \leftrightarrow R_2$ $\sim \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 2\\ 0 & 3\end{bmatrix}$

$R_3 \leftarrow R_3 - 2R_2$

$R_4 \leftarrow R_4 - 3R_2$ $\sim \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\\ 0 & 0\end{bmatrix}$

Hence, rank $(A) = 2$

Hence, $\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$ are linearly independent

and since, $V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}$,

thus, $\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}$ form a basis for V

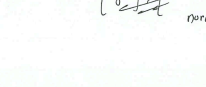
~~Notice $\vec{u}_1 \cdot \vec{u}_2 = 0$, thus orthogonal~~

by Gram-Schmidt orthogonalization. $\vec{u}_1' = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$,

$\vec{u}_2' = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{\begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\cdot\begin{bmatrix}0\\1\\1\\0\end{bmatrix}}{\left\|\begin{bmatrix}0\\1\\1\\0\end{bmatrix}\right\|^2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-\sqrt{2}\\2-\sqrt{2}\\-1\end{bmatrix}$

$\sqrt{2}$ ~~$\sqrt{2}$~~ 2 ~~1~~

~~Hence, orthonormal basis for $V = \{\vec{u}_1', \vec{u}_2'\}$~~ ~~notice $\vec{u}_1' \cdot \vec{u}_2' \neq 0$~~



normalize $\vec{u}_1'$.

$\vec{u}_1' = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}$

normalize $\vec{u}_2'$ = $\begin{bmatrix}\frac{3}{\sqrt{14}}\\-\frac{1}{\sqrt{14}}\\\frac{1}{\sqrt{14}}\\-\frac{1}{\sqrt{14}}\end{bmatrix}$

Hence, orthonormal basis for V is

$\{\vec{u}_1', \vec{u}_2'\} = \left\{\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \begin{bmatrix}\frac{3}{\sqrt{14}}\\-\frac{1}{\sqrt{14}}\\\frac{1}{\sqrt{14}}\\-\frac{1}{\sqrt{14}}\end{bmatrix}\right\}$

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

pf:

$$A^T = -A$$

$$\det(A^T) = \det(-A)$$

for $n \times n$ matrix, $\det(-A) = (-1)^n \det(A)$

$$\det(A^T) = (-1)^n \det(A)$$

Since $\det(A^T) = \det(A)$

$$\det(A) = (-1)^n \det(A)$$

if n is odd, then $(-1)^n = -1$

$$\det(A) = -\det(A)$$

$$\det(A) = 0 \quad \square$$

3/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

pf:

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$$

$$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$$

$$(\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v}) = (\vec{u}-\vec{v})\cdot(\vec{u}-\vec{v})$$

$$\|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\vec{u}\cdot\vec{v} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\vec{u}\cdot\vec{v}$$

$$\vec{u}\cdot\vec{v} = 0$$

$$\vec{u} \perp \vec{v} \quad \square$$